

Intelligent Systems

Introduction



About the course

Main topics

- Solving problems by searching
- Knowledge and reasoning
- Machine learning
- Language processing (optional)

Literature:

- Ivan Bratko. Prolog Programming for Artificial Intelligence (4th Edition). Pearson, 2011.
- Adrian A. Hopgood. Intelligent Systems for Engineers and Scientists (3rd Edition). CRC Press, 2011.
- I. Kononenko in M. Robnik Šikonja: Inteligentni sistemi. Založba FE in FRI, 2010.
- Stuart Russell in Peter Norvig. Artificial Intelligence: A Modern Approach (4th Edition). Prentice Hall, 2021.

About the course

Student's obligations:

- 2 programming assignments + presentations [40%]
 - Search & reasoning (middle of semester) [20%]
 - Machine learning (end of semester) [20%]
- Written exam + oral exam [60%]
- Oral exam (optional)

Written exam can be substituted with 2 midterm exams, each at least 50% (15% of the final grade).

The student must complete and present both programming assignments.

What is AI?



Definition of AI

“The science and engineering of making intelligent machines”

— John McCarthy (1955)

“AI is the study of ideas which enable computers to do the things that make people seem intelligent”

— Patrick Winston (1979)

“The study and design of intelligent agents, where an intelligent agent is a system that perceives its environment and takes actions that maximize its chances of success.”

— Russell & Norvig

The problem with the definition

What is (human) intelligence?

- The ability to reason, solve problems?
- The ability to learn?
- The ability to communicate ideas?
- Creativity, emotions, consciousness?
- ...

Which of the above are the most difficult (if even possible)?

Philosophical views

Weak AI

- The digital computer is a useful tool for studying intelligence and developing useful technology.
- A running AI program a simulation of a cognitive process but is not itself a cognitive process.
- The machines can be made to act as if they are intelligent.

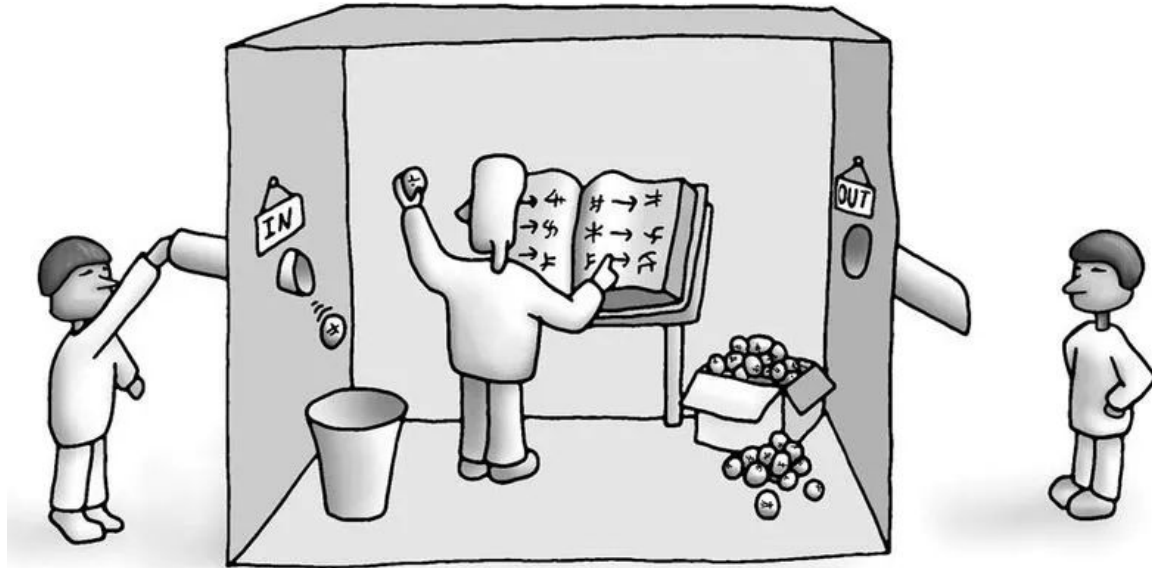
Philosophical views

Strong AI

- The digital computer can actually be programmed to be a mind.
- Replicate human cognitive abilities, including consciousness, self-awareness, and intentionality (AGI).
- Currently no true strong AI exists.

Philosophical views

The Chinese Room Argument (John Searle, 1980)



Philosophical views

Responses to the Chinese room argument

- The whole system (including the room) understands.

Searle: Let the person internalize the entire system.

- If the program was simulating the brain, it would understand.

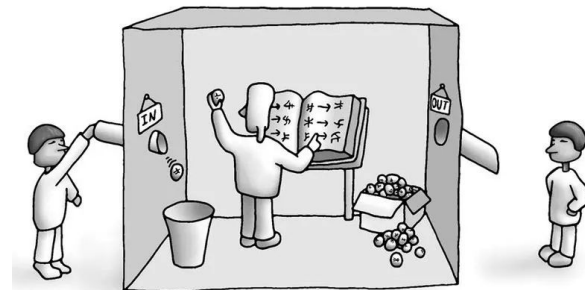
Searle: Simulation is not duplication. Simulated brain would lack causal power for consciousness.

- Maybe not a digital computer, but a neurons-like device?

Searle: This admits the failure of strong AI.

- We can't know if even other people really have minds.

Searle: We infer other humans have minds because they are biologically similar to us.

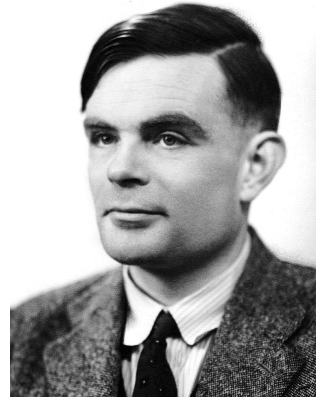


Alan Turing (1912 - 1954)

Can machines think?

The argument from disability. A machine can never:

- Have initiative?
- Make mistakes?
- Have a sense of humor?
- Tell right from wrong?
- Fall in love?
- Be the subject of its own thought?
- Do something really new?



Gödel's Incompleteness Theorem

Gödel (1936) and Turing (1936) proved, that certain mathematical questions are unanswerable by particular formal systems.

Gödel's Theorem

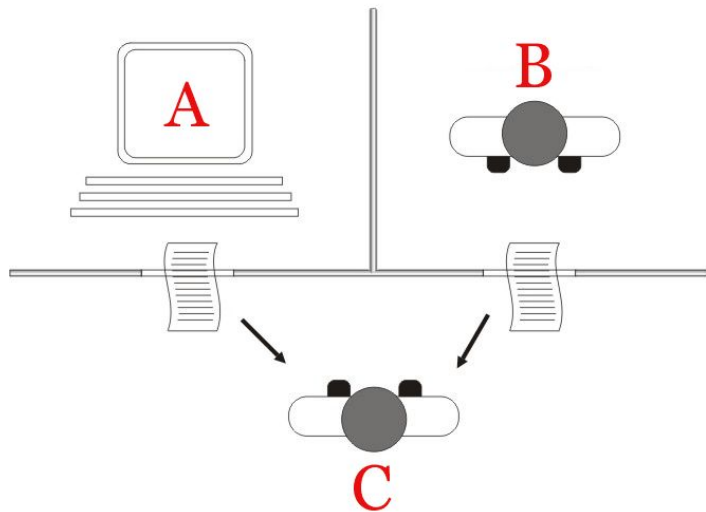
For any formal axiomatic framework F powerful enough to do arithmetic, it is possible to construct a sentence $G(F)$ with properties:

- $G(F)$ is a sentence of F , but cannot be proved in F .
- If F is consistent, then $G(F)$ is true.

Machines are formal systems that are limited to by the incompleteness theorem — J. R. Lucas (1961)

Turing's Test (1950)

Instead of asking whether machines can think, ask if they can pass a behavioral test.



C. R. Jones, et al.:

Large Language Models Pass the Turing Test, March 2025

- Participants engaged in simultaneous 5-minute conversations with another human participant and one of these systems.
- Asked which partner was a human.
- GPT-4.5 was identified as the human 73% of the time—significantly more often than the actual human participants.

Four schools of thought

BEHAVIOR	Acting humanly	Acting rationally
	The Turing test approach • natural language processing • knowledge representation • automated reasoning • machine learning • computer vision • robotics How to act as a human?	The rational agent approach • Acting to achieve best outcome • Optimization methods • May involve “thinking rationally” • Control theory How to achieve goals?
THOUGHT	Thinking humanly	Thinking rationally
	The cognitive modeling approach • introspection • psychological experiments • brain imaging How to think as a human?	The “laws of thought” approach • Formal systems of deduction • Predictions about the future • The theory of probability How to think correctly?

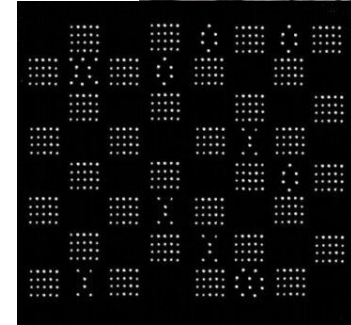
HUMANRATIONAL

Brief history AI?



The inception of AI (1943 - 1956)

- 1943, Warren McCulloch & Walter Pitts
A model of artificial neurons (on/off), logical connections (AND, OR, ...).
- 1950, Marvin Minsky & Dean Edmonds
The first neural network computer (40 neurons).
- 1952, Christopher Strachey, Arthur Samuel
A program playing checkers.
- 1956 John McCarthy & colleagues
Two-month workshop, 10 attendees.
Novell & Simon → A theorem proving program.



Early enthusiasm (1952 - 1969)

- 1950s, AI researchers focus on solving puzzles, games, mathematics, IQ tests.
- 1959, Herbert Gelernter
Geometry Theorem Prover - precursor of modern mathematical theorem provers.
- 1956, Arthur Samuel
Used reinforcement learning, the program was able to defeat it's creator.
- 1958, John McCarthy
The List programming language.
- 1960s, early work on neural networks.

A dose of reality (1966 - 1973)

- Predictions:
Within 10 years → AI chess championship, significant mathematical theorems proven, ...
- Problems:
Trying to mimic humans, not yet understanding computational complexity.
- First attempts at genetic algorithms:
Thousands of hours of CPU time without significant progress.
- It was proven that perceptrons (simple ANN) can learn anything (1969),
but there was little progress in practice.

Expert systems (1969 - 1986)

- Knowledge base + specialized inference engine + UI.
- DENDRAL (1969), chemical analysis and molecular structure identification.
- MYCIN (1970s), medical diagnosis system for bacterial infections.
- Prolog programming language becomes popular.
- Difficult to build and maintain expert systems for complex domain.
- Expert systems cannot learn from experience.

The return of neural networks (1986 - present)

- Reinvention of the back-propagation algorithm.
- Competitors to the symbolic and logistic AI.
- The capability to learn from examples.
- Humans reason symbolically. But is this really needed in AI?

Machine learning (1987 - present)

- Moving to real-world applications.
- Shared benchmark problems (UC Irvine Machine Learning Repository).
- Speech recognition (LibriSpeech), image recognition (MNIST, ImageNet).
- Probabilistic reasoning: Bayesian Networks.
- Markov Decision process, reinforcement learning.
- Computer vision.
- Robotics.

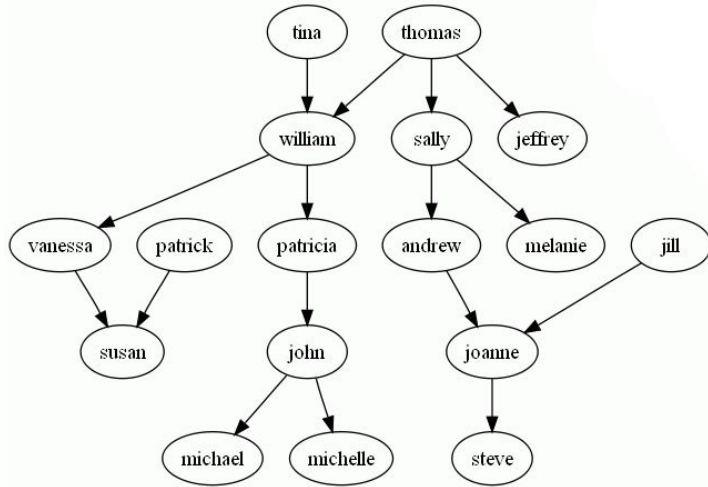
Deep learning (2011 - present)

- Multiple layers of computing elements.
- First success in speech recognition, then computer vision.
- Convolutional neural networks.
- Winning the game Go (2015).
- Recommender systems.
- Enabled by Big data and specialized hardware (GPU, TPU, ...).
- Generative Models, Large Language Models, ...

Examples of AI applications



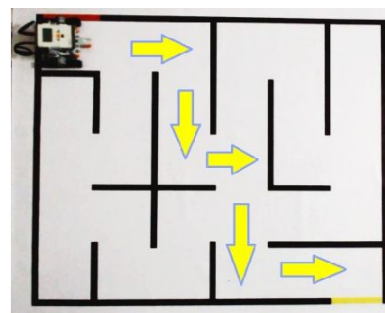
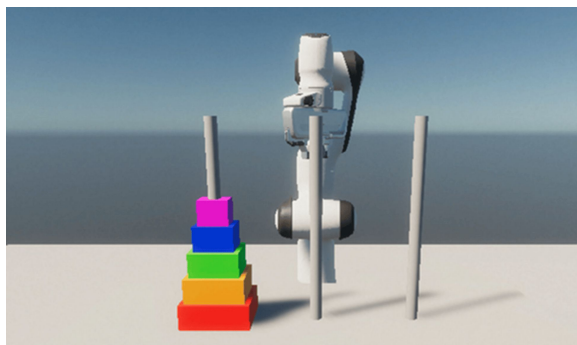
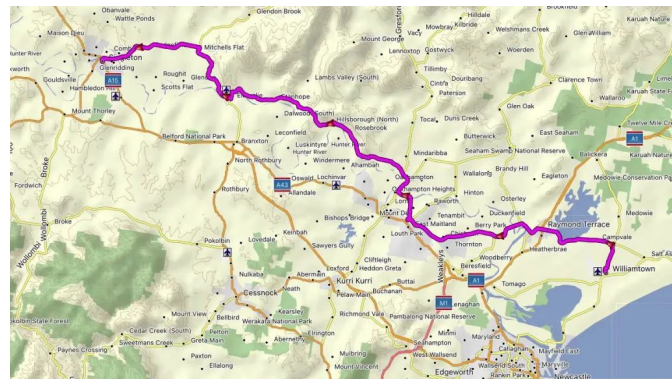
Automated Reasoning (Prolog)



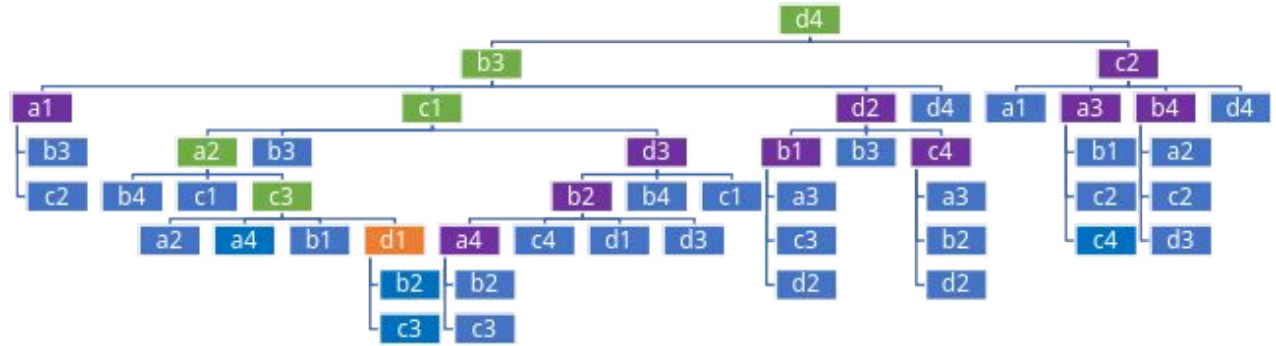
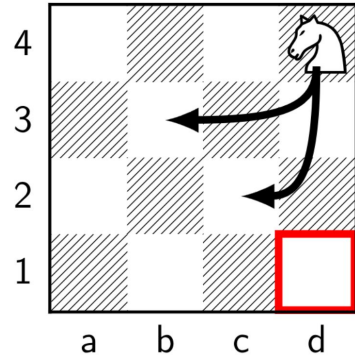
```
parent(tina,william).
parent(thomas,william).
parent(thomas,sally).
...
male(william).
male(thomas).
female(tina).
female(sally).
...
sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.
aunt(X, Y) :- parent(P, Y), sibling(P, X), female(X).

? aunt(X, susan)
X = patricia
```


Search and Planning



Games & puzzles



Shortest solution: $d4 \rightarrow b3 \rightarrow c1 \rightarrow a2 \rightarrow c3 \rightarrow d1$

Deep learning

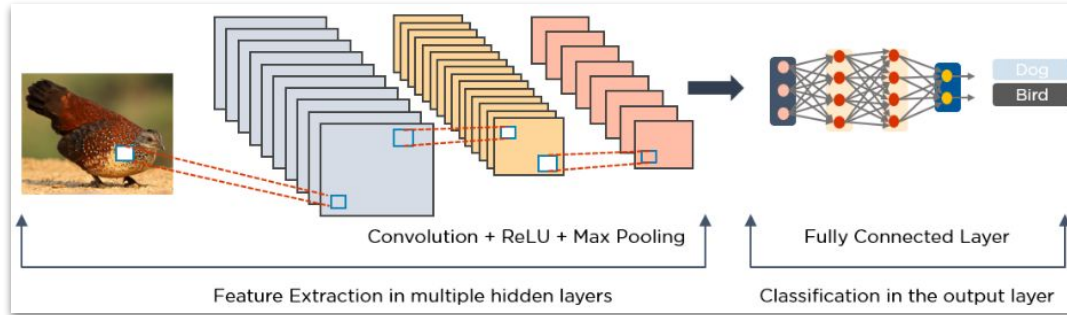
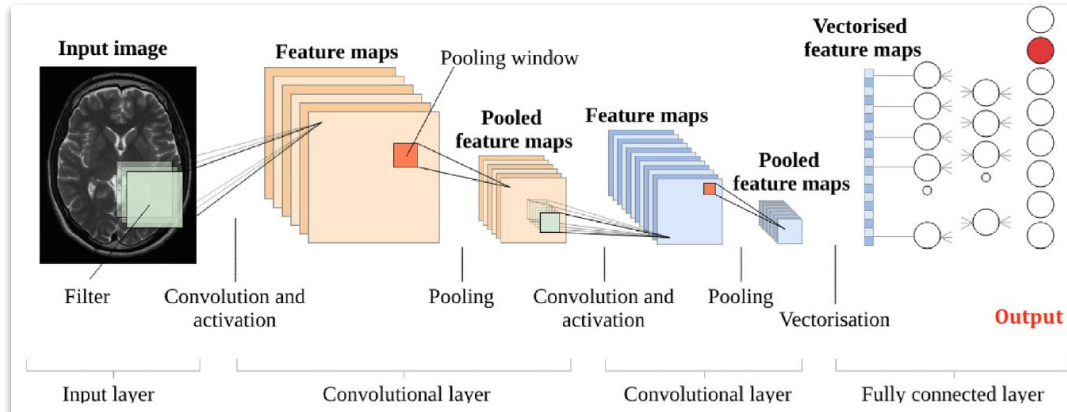
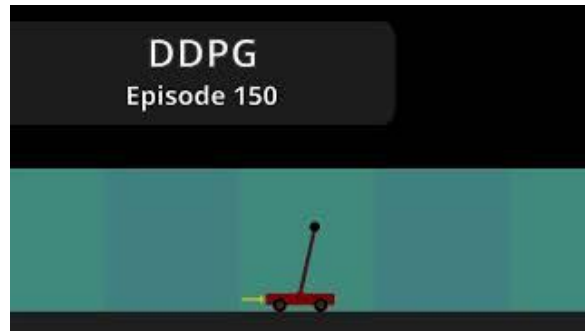
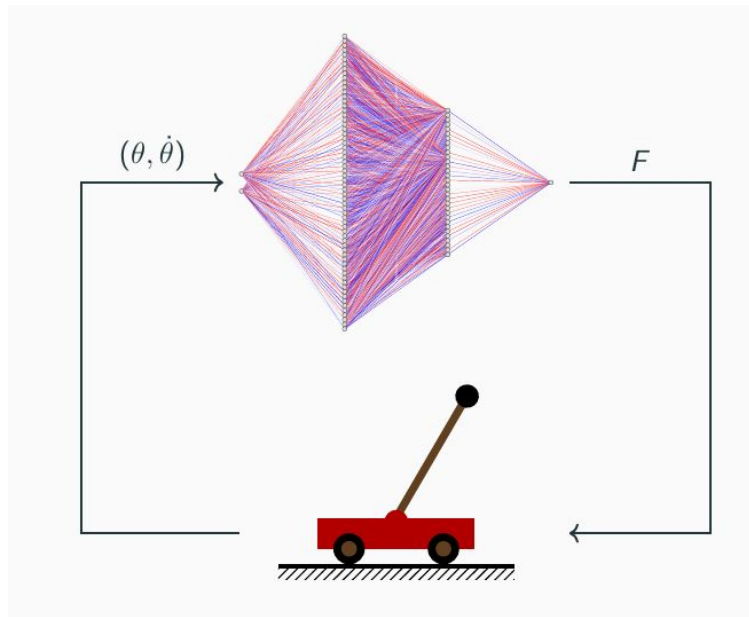


Image recognition

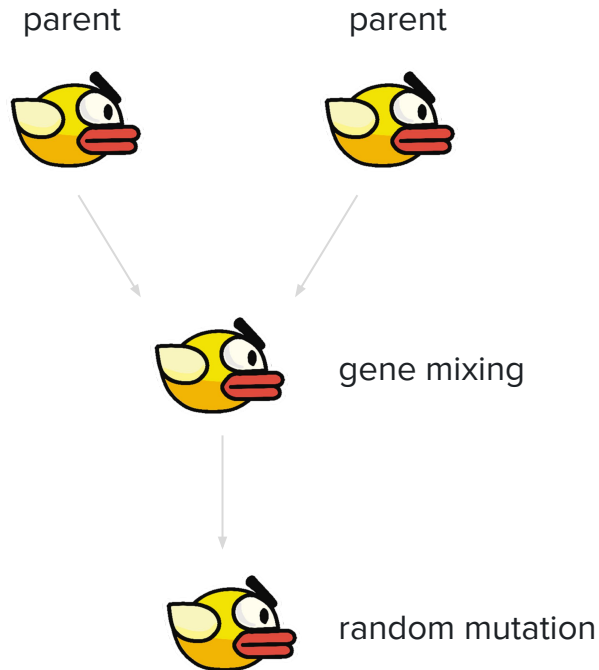


Medical diagnostics

Reinforcement Learning



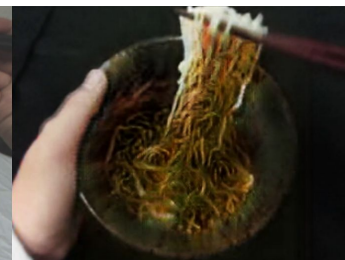
Genetic algorithms



AI agents



Generative AI



Style Transfer



Summer $\rightleftharpoons$ Winter



summer $\rightarrow$ winter



winter $\rightarrow$ summer